Lab 7

Fuzzing for Input Validation Bugs

L6_L7_42 - Salon Booking

Tool: Hypothesis (property-based fuzz testing for Python)

Repository: github.com/akshat-k30/SE-Lab6-L6_L7_42_salon_booking

Step 2: Run the Buggy Application

Command: python salon_booking.py

The buggy code runs fine with well-formed inputs but has NO input validation. It will crash on None, empty strings, missing delimiters, or non-numeric values.

```
PS> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): Service and Duration: ('Haircut', 45.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): Average Duration: 45

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): Payable Bill: 844.0
```

Step 4: Run Fuzz Tests Against Buggy Code - ALL FAILING

Command: pytest test_salon_booking.py -v

Hypothesis generates random inputs (including None). All 3 tests FAIL with AttributeError because the buggy functions call .split() on None.

Note: The full pytest output is very long; the screenshot below shows the condensed key lines (header, FAILED markers, error types, and summary).

```
PS> pytest test_salon_booking.py -v
============================= test session starts =============================
platform win32 -- Python 3.13.14, pytest-9.2.0, pluggy-1.7.0 --
    C:\Users\aksha\AppData\Local\Programs\Python\Python313\python.exe
hypothesis profile 'default'
plugins: anyio-4.14.2, hypothesis-6.170.15
collecting ... collected 3 items

test_salon_booking.py::test_parse_booking_no_crash FAILED                [ 33%]
test_salon_booking.py::test_duration_average_no_crash FAILED             [ 66%]
test_salon_booking.py::test_payable_bill_no_crash FAILED                 [100%]

================================== FAILURES ===================================
_____________________________ test_parse_booking_no_crash _____________________________

s = None

E       AttributeError: 'NoneType' object has no attribute 'split'

s = None

E           AssertionError: parse_booking crashed with AttributeError: 'NoneType' object has no attribute
    'split' on input: None

test_salon_booking.py:27: AssertionError
_____________________________ test_duration_average_no_crash _____________________________

s = None

E       AttributeError: 'NoneType' object has no attribute 'split'

s = None

E           AssertionError: duration_average crashed with AttributeError: 'NoneType' object has no
    attribute 'split' on input: None

test_salon_booking.py:52: AssertionError
_____________________________ test_payable_bill_no_crash _____________________________

s = None

E       AttributeError: 'NoneType' object has no attribute 'split'

s = None

E           AssertionError: payable_bill crashed with AttributeError: 'NoneType' object has no attribute
    'split' on input: None

test_salon_booking.py:76: AssertionError

=========================== short test summary info ===========================
FAILED test_salon_booking.py::test_parse_booking_no_crash - AssertionError: p...
FAILED test_salon_booking.py::test_duration_average_no_crash - AssertionError...
FAILED test_salon_booking.py::test_payable_bill_no_crash - AssertionError: pa...
======================== 3 failed, 1 warning in 1.16s =========================
```

Step 5: Run the Fixed Application

Command: python salon_booking.py

All three functions have been hardened with type checks, delimiter validation, empty-value detection, numeric parsing guards, and range validation. The fixed code produces the same correct output for valid inputs.

```
PS> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): Service and Duration: ('Haircut', 45.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): Average Duration: 45

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): Payable Bill: 844.0
```

Step 6: Re-Run Fuzz Tests Against Fixed Code - ALL PASSING

Command: pytest test_salon_booking.py -v

All 15 Hypothesis fuzz tests now PASS. The test suite includes 3 property-based fuzz tests (one per function) plus 12 explicit edge-case tests covering negative durations, NaN, infinity, and valid-input verification.

```
PS> pytest test_salon_booking.py -v
============================= test session starts =============================
platform win32 -- Python 3.13.14, pytest-9.2.0, pluggy-1.7.0 --
    C:\Users\aksha\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\developer\labs\se_lab\labs\SE-Lab6-Lab7\lab6\SE-Lab6-L6_L7_42_salon_booking\Lab 7
plugins: anyio-4.14.2, hypothesis-6.170.15
collecting ... collected 15 items

test_salon_booking.py::test_parse_booking_no_crash PASSED                [  6%]
test_salon_booking.py::test_parse_booking_rejects_nan PASSED             [ 13%]
test_salon_booking.py::test_parse_booking_rejects_inf PASSED             [ 20%]
test_salon_booking.py::test_parse_booking_rejects_negative_inf PASSED    [ 26%]
test_salon_booking.py::test_parse_booking_valid PASSED                   [ 33%]
test_salon_booking.py::test_duration_average_no_crash PASSED             [ 40%]
test_salon_booking.py::test_duration_average_rejects_negative PASSED     [ 46%]
test_salon_booking.py::test_duration_average_rejects_all_negative PASSED [ 53%]
test_salon_booking.py::test_duration_average_valid PASSED                [ 60%]
test_salon_booking.py::test_payable_bill_no_crash PASSED                 [ 66%]
test_salon_booking.py::test_payable_bill_rejects_nan_bill PASSED         [ 73%]
test_salon_booking.py::test_payable_bill_rejects_inf_bill PASSED         [ 80%]
test_salon_booking.py::test_payable_bill_rejects_nan_discount PASSED     [ 86%]
test_salon_booking.py::test_payable_bill_rejects_inf_discount PASSED     [ 93%]
test_salon_booking.py::test_payable_bill_valid PASSED                    [100%]

============================== warnings summary ===============================
..\..\..\..\..\..\..\Users\aksha\AppData\Local\Programs\Python\Python313\Lib\site-
    packages\anyio\pytest_plugin.py:14
  C:\Users\aksha\AppData\Local\Programs\Python\Python313\Lib\site-packages\anyio\pytest_plugin.py:14:
    PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 15 passed, 1 warning in 1.00s ========================
```

Summary of Bugs Found and Fixes Applied

Bugs Discovered via Fuzz Testing:

1. parse_booking(None) --> AttributeError (no null check)
2. parse_booking('haircut') --> ValueError (missing ':' delimiter)
3. parse_booking('haircut:abc') --> ValueError (non-numeric duration)
4. parse_booking(':45') --> returns empty service name
5. parse_booking('haircut:nan') --> accepts NaN as valid duration
6. parse_booking('haircut:inf') --> accepts infinity as valid duration
7. duration_average(None) --> AttributeError (no null check)
8. duration_average('') --> ZeroDivisionError (empty list, div by zero)
9. duration_average('30,,60') --> ValueError (empty CSV element)
10. duration_average('30,abc,60') --> ValueError (non-integer value)
11. duration_average('30,-5,60') --> accepts negative durations
12. payable_bill(None) --> AttributeError (no null check)
13. payable_bill('800') --> ValueError (missing '|' delimiter)
14. payable_bill('abc|100') --> ValueError (non-numeric bill)
15. payable_bill('nan|10') --> accepts NaN as valid bill
16. payable_bill('800|inf') --> accepts infinity as valid discount

Fixes Applied (Defensive Programming):

- Type checking: reject None and non-string inputs with ValueError
- Delimiter validation: check for required separators (':', ',', '|')
- Empty value detection: reject empty fields after splitting
- Numeric parsing guards: try/except around int()/float() conversions
- Finite number checks: reject NaN, inf, -inf using math.isfinite()
- Range validation: reject negative durations, negative bills/discounts
- Graceful error handling in main() with try/except for user input

Test Suite Summary:

15 total tests: 3 Hypothesis property-based fuzz tests (one per function) plus 12 explicit edge-case tests verifying rejection of NaN, infinity, negative values, and correct results for valid inputs. All 15 tests pass - the code handles every edge case gracefully.